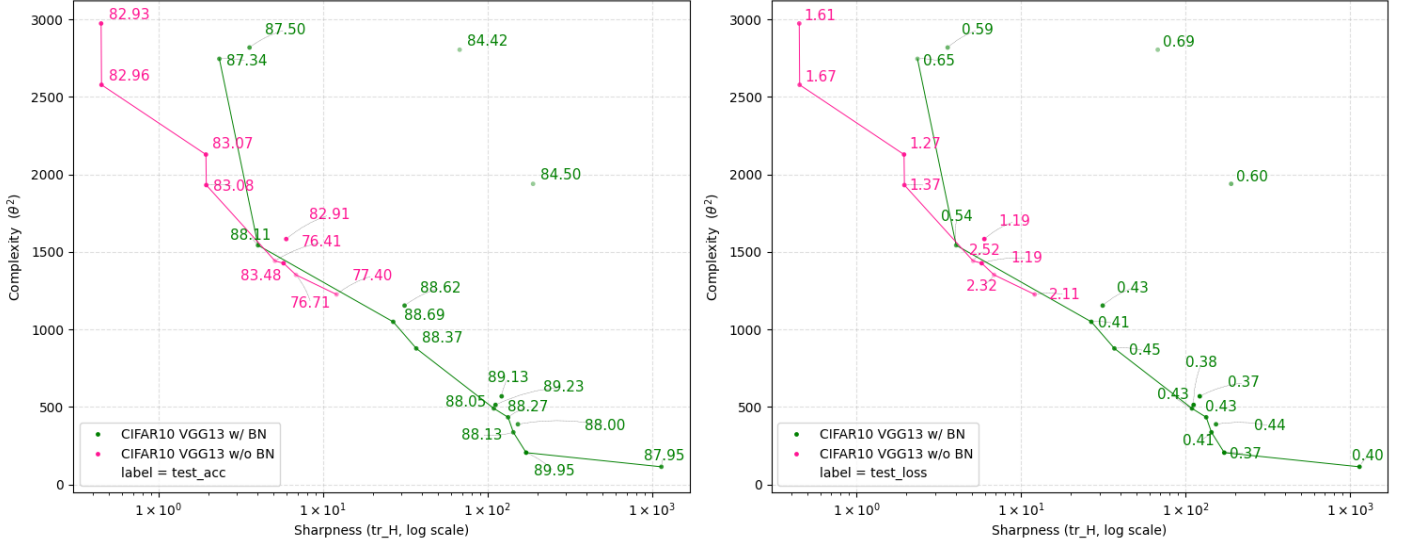
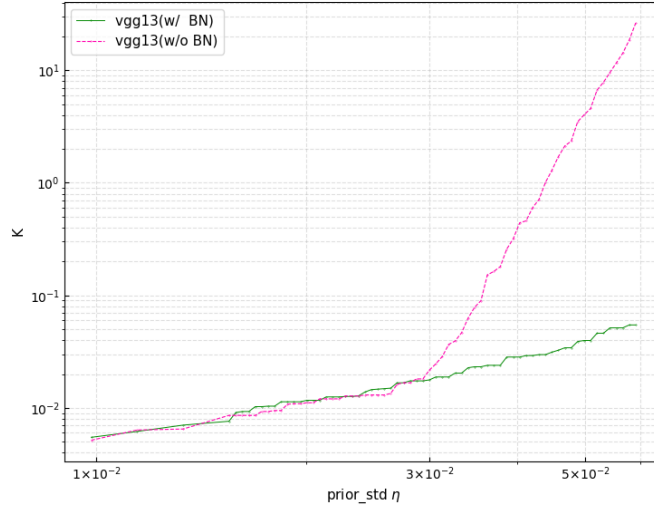


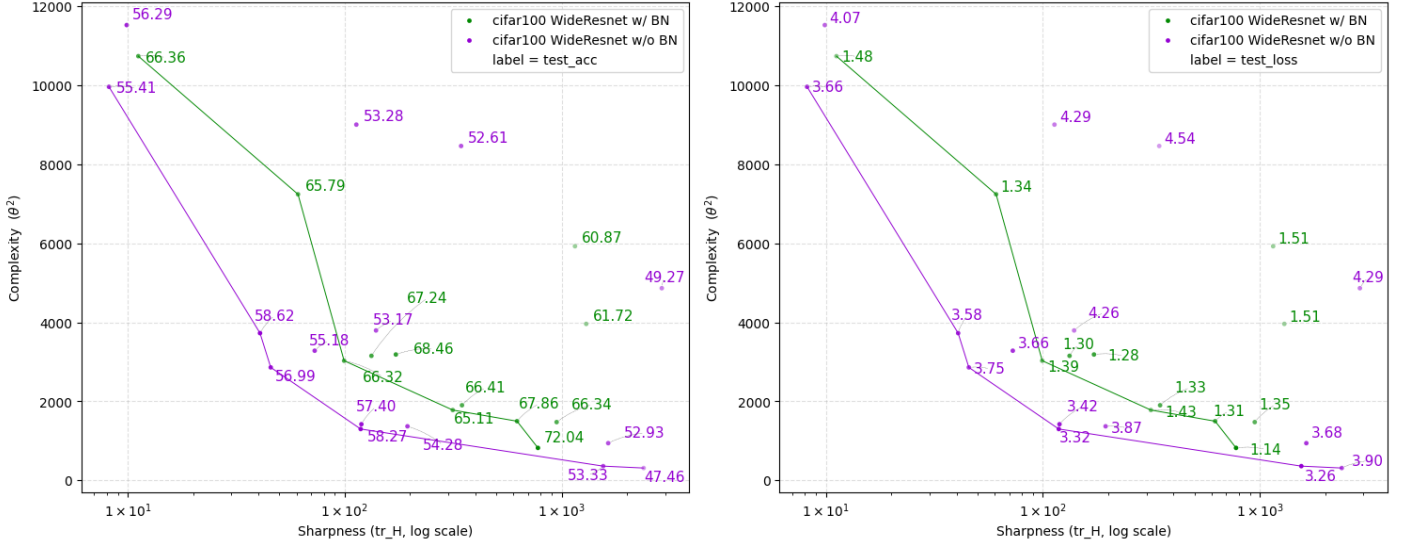


(a) Empirical Pareto frontiers (S, KL) of VGG-13 with batch norm (*green*) and without batch norm (*pink*). Sharpness S is estimated by trace of the Hessian. Complexity KL is estimated by $\|\theta\|^2$. Each point represents a hyperparameter setting; settings that meet train loss ≤ 0.01 are kept and plotted. The text annotations denote the generalization performance (left is test accuracy, higher is better; right is test loss, lower is better).

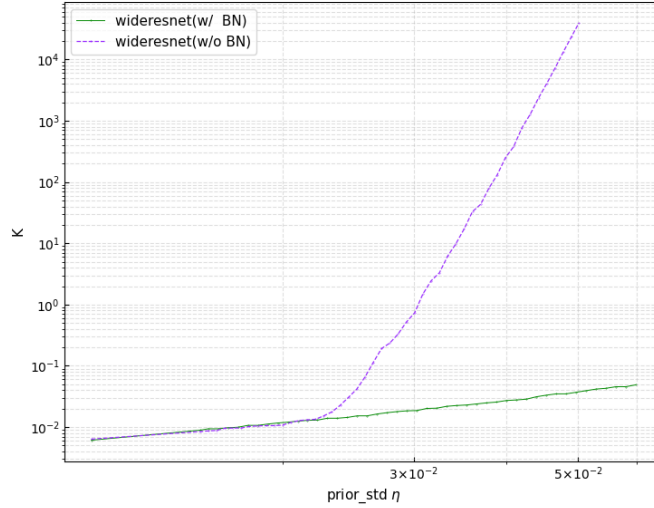


(b) K values of VGG-13 with batch norm (*green*) and without batch norm (*pink*) under prior $\mathcal{N}(0, \eta^2 \cdot I)$, $\eta = \{0.01, \dots, 0.06\}$. Removing batch norm inflates K .

Figure 2: Illustration of the three-factor framework using VGG-13 on CIFAR-10. Models trained without BN (*pink*) yield fewer converged points due to optimization difficulties. Removing BN degrades generalization and inflates K . Though some achieve smaller S and KL relative to those with BN, their worse performance is explained by the inflated K , supporting our framework. Furthermore, the (S, KL) Pareto curve without BN exhibits internal inconsistencies: some *pink* points achieve smaller S and KL , yet yield worse test loss than others (e.g., 2.52 versus 1.19). This shows that a massive K fundamentally loosens the bound, rendering the (S, KL) Pareto curve unreliable, and indicating that the chosen prior is inappropriate for no-BN VGG-13 architecture. This manifests K 's **practical role** as an essential factor accounting for generalization as well as a diagnostic indicator of the reliability of the (S, KL)-plane comparison and the quality of the prior.



(a) Empirical Pareto frontiers (S, KL) of WideResnet with batch norm (*green*) and without batch norm (*purple*). Sharpness S is estimated by trace of the Hessian. Complexity KL is estimated by $\|\theta\|^2$. Each point represents a hyperparameter setting; settings that meet train loss ≤ 0.01 are kept and plotted. The text annotations denote the generalization performance (left is test accuracy, higher is better; right is test loss, lower is better).



(b) K values of WideResnet with batch norm (*green*) and without batch norm (*purple*) under prior $\mathcal{N}(0, \eta^2 I)$, $\eta = \{0.01, \dots, 0.06\}$. Removing batch norm inflates K .

Figure 3: Illustration of the three-factor framework using WideResnet on CIFAR-100. Without batch norm (*purple*) achieves both smaller S and KL , i.e., frontier shifts lower-leftward, yet exhibits worse generalization performance. The discrepancy is resolved when incorporating its inflated K . This manifests K 's **practical role** that it captures variations that cannot be explained by (S, KL) .